APIM AI Gateway — Multi-Foundry Patterns

Routing, Load Balancing, Quota & Throttling for Multi-Model and Multi-Region Foundry Deployments

2026

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Scale **multiple Azure AI Foundry resources** behind a single **Azure API Management (APIM) AI Gateway** — multi-region failover, weighted load balancing, semantic routing, centralized infrastructure-as-code, and resilient quota handling for streaming and agent-to-agent workloads.

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1 Overview

The companion document Foundry_Agent_Monitoring_APIM describes how a **single** Foundry resource can be exposed via APIM for per-user token tracking. This guide answers the next question: **what happens when one Foundry instance is no longer enough?**

Common scaling triggers:

- **Quota exhaustion** — a single AOAI / Foundry deployment hits its TPM (tokens-per-minute) ceiling per region.
- **Multi-region resiliency** — fail over to another region when the primary is throttled or unavailable.
- **Multi-model exposure** — expose `gpt-4o`, `gpt-4o-mini`, `o4-mini`, `text-embedding-3-large`, etc. through one stable URL.
- **Cost optimization** — route cheap prompts to a cheap model, expensive prompts to a frontier model.

- **Tenant / team isolation** — different APIM Products consume different backend pools.

APIM acting as an **AI Gateway** solves all of these without requiring any change in the calling application — clients keep talking to a single URL.

> **Foundry context** — A "Foundry resource" exposes its inference endpoint at `https://<resource>.services.ai.azure.com/api/projects/<project>`. Inside, deployments can be Azure OpenAI, Azure AI Inference, Cohere, Mistral, Llama, or any model from the catalog. APIM treats each one as a backend.

1.1 What APIM AI Gateway adds

Capability	Benefit
Backend Pool (GA, October 2024)	Round-robin / weighted / priority load balancing across N backends
Circuit Breaker	Auto-eviction on `429` / `5xx` for a configurable cooldown
`azure-openai-token-limit` **policy**	TPM enforcement per subscription / user / IP
`azure-openai-emit-token-metric` **policy**	Per-caller token + cost metrics in Application Insights
`llm-semantic-cache-lookup/store`	Embedding-based response cache (cuts cost ~30-50% on FAQ workloads)
`llm-semantic-routing` **(preview)**	Route by **prompt meaning**, not by URL or header
Managed Identity	No keys to rotate — APIM authenticates to Foundry with its system-assigned identity

2 Architecture

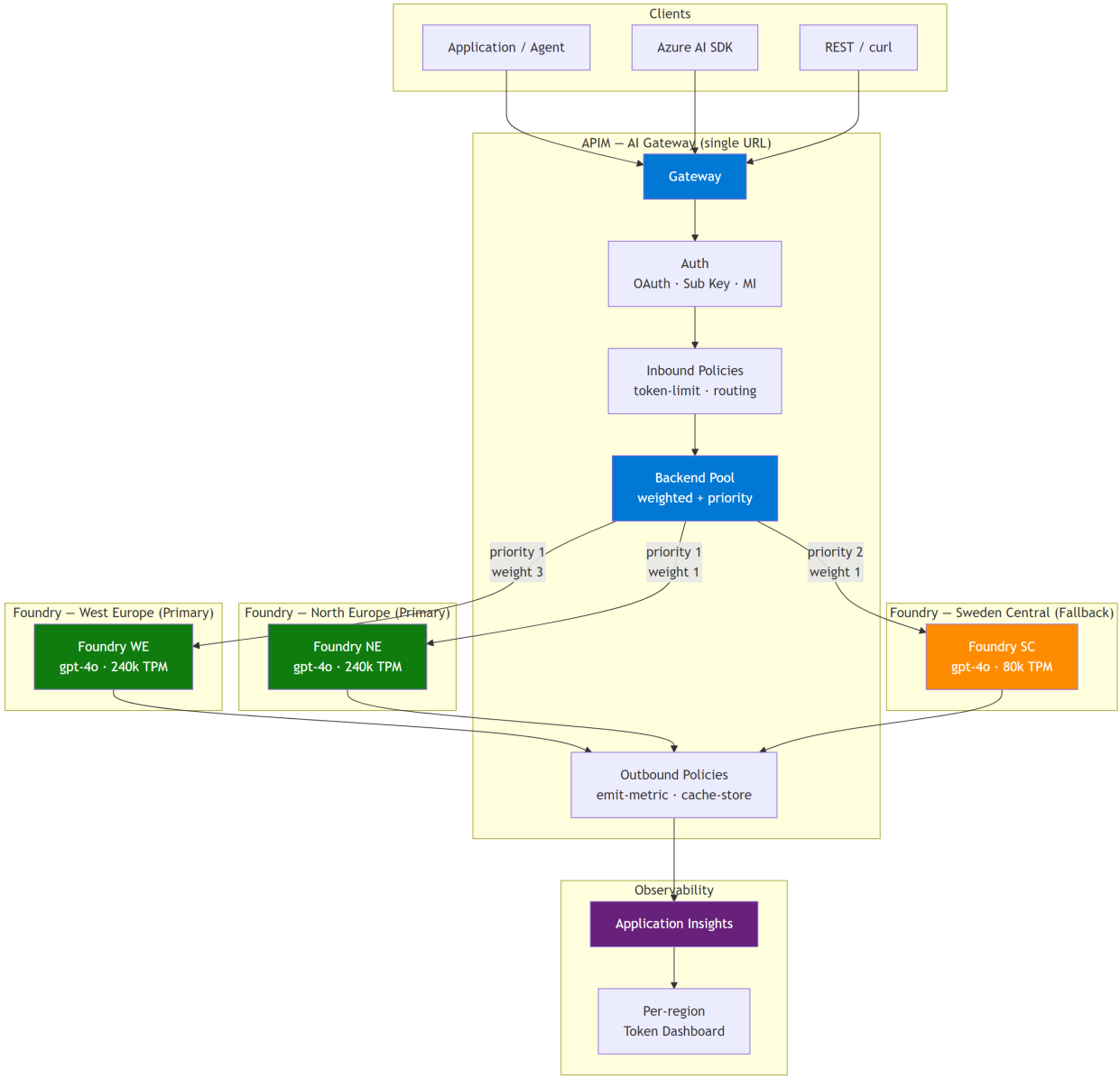

Figure 1: Diagram 1

Key idea — clients see one URL. APIM owns the routing logic, the keys, the metrics, and the resilience. Foundry resources can be added or removed without changing client code.

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3 Security & Governance

A multi-Foundry gateway is also a **single point of policy enforcement**. Centralizing security and governance there avoids re-implementing them in every client.

3.1 Authentication — kill the API keys

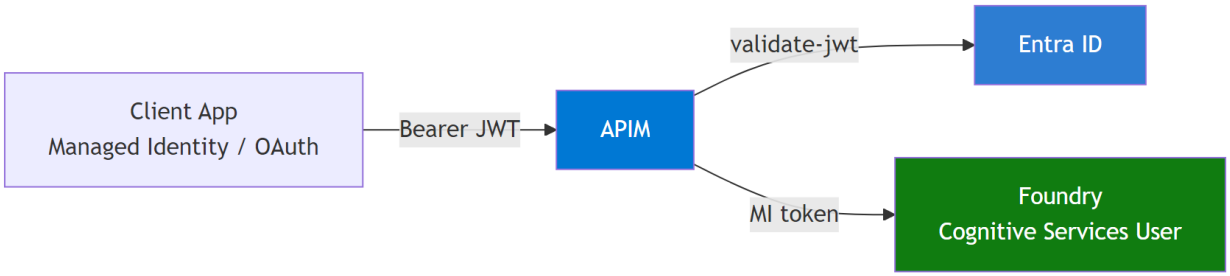

Figure 2: Diagram 2

Hop	Recommended	Avoid
Client → APIM	OAuth2 / OIDC bearer token (Entra ID) — validated by `<validate-jwt>`	Long-lived `Ocp-Apim-Subscription-Key` in source code
APIM → Foundry	System-assigned **Managed Identity** + RBAC `Cognitive Services User`	Foundry API keys stored in named-values
Service-to-service	Workload identity federation (GitHub / Argo / AKS)	Service principal client secret

Inbound policy:

```
<inbound>
  <base />
  <validate-jwt header-name="Authorization"
↪   failed-validation-httpcode="401">
    <openid-config url="https://login.microsoftonline.com/{{tenant-
↪   id}}/v2.0/.well-known/openid-configuration" />
    <required-claims>
      <claim name="aud"><value>{{api-app-id}}</value></claim>
    </required-claims>
  </validate-jwt>
  <set-backend-service backend-id="foundry-pool" />
  <authentication-managed-identity
↪   resource="https://cognitiveservices.azure.com" />
</inbound>
```

3.2 Content Safety — filter prompts and completions

Add the `llm-content-safety` policy to forward every prompt (and optionally completion) to **Azure AI Content Safety** before it reaches the model. Requests that exceed the configured severity thresholds are rejected at the gateway with `403`, never billed to Foundry.

```
<inbound>
  <base />
  <llm-content-safety
      backend-id="content-safety"
      shield-prompt="true">
    <categories>
      <category name="Hate"     threshold="4" />
      <category name="Violence" threshold="4" />
      <category name="Sexual"   threshold="4" />
      <category name="SelfHarm" threshold="4" />
    </categories>
  </llm-content-safety>
</inbound>
```

> **Prompt-shield** also catches indirect prompt injection (poisoned RAG documents, untrusted tool outputs). Recommended on any agent that can call tools or fetch external content.

3.3 Governance — quotas, products, attribution

APIM **Products** are the unit of governance. Each consuming team subscribes to a Product, which carries its own subscription key, quota, and allowed APIs.

```
<inbound>
  <base />
  <llm-token-limit
      counter-key="@(context.Subscription.Id)"
      tokens-per-minute="60000"
      tokens-per-day="2000000"
      estimate-prompt-tokens="true"
      tokens-consumed-header-name="x-tokens-consumed"
      remaining-tokens-header-name="x-tokens-remaining" />
</inbound>
```

Lever	What it gives you
`llm-token-limit` per `context.Subscription.Id`	Hard TPM/TPD ceiling per team
Per-Product policy	Different quotas for `team-marketing` vs `team-research`
`azure-openai-emit-token-metric` with `user-id` dimension	Per-user attribution in App Insights

Lever	What it gives you
APIM Developer Portal	Self-service onboarding — teams request access, get a key, view their quota
Subscription approval workflow	Manual gate for sensitive Products (e.g. legal, HR)

Cross-link — per-user / per-agent attribution mechanics (KQL queries, dashboards) are detailed in Foundry_Agent_Monitoring_APIM.

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4 Pattern 1 — Multi-Endpoint Exposure (Named Backends)

The simplest pattern. Each model or region is a **named backend**; the routing decision is made by reading a request property (header, query parameter, or body field).

4.1 Backend definition (Bicep)

```
resource backendGpt4o 'Microsoft.ApiManagement/service/backends@2024-05-01'
 ↪  = {
  name: 'foundry-gpt-4o'
  parent: apim
  properties: {
    url:
 ↪  'https://foundry-we.services.ai.azure.com/openai/deployments/gpt-4o'
    protocol: 'http'
    credentials: {
      header: { 'api-key': [ '{{foundry-we-key}}' ] }
    }
  }
}

resource backendGpt4oMini
 ↪  'Microsoft.ApiManagement/service/backends@2024-05-01' = {
  name: 'foundry-gpt-4o-mini'
  parent: apim
  properties: {
    url: 'https://foundry-we.services.ai.azure.com/openai/deployments/gpt-
 ↪  4o-mini'
    protocol: 'http'
  }
}
```

4.2 Routing policy (header-based)

```
<policies>
  <inbound>
    <base />
    <choose>
      <when condition="@(context.Request.Headers.GetValueOrDefault("x-model
↪  ","").Equals("gpt-4o-mini"))">
        <set-backend-service backend-id="foundry-gpt-4o-mini" />
      </when>
      <when condition="@(context.Request.Headers.GetValueOrDefault("x-model
↪  ","").Equals("gpt-4o"))">
        <set-backend-service backend-id="foundry-gpt-4o" />
      </when>
      <otherwise>
        <set-backend-service backend-id="foundry-gpt-4o-mini" />
      </otherwise>
    </choose>
  </inbound>
</policies>
```

When to use — small number of models, deterministic routing, no resiliency requirements. Each backend is a single point of failure.

---

5 Pattern 2 — Backend Pool with Load Balancing & Circuit Breaker

The recommended pattern for **resilient multi-region** deployments. APIM's `Pool` backend type combines several backends into one logical target; APIM automatically distributes traffic according to **priority** and **weight** and ejects unhealthy backends via a **circuit breaker**.

5.1 Concept

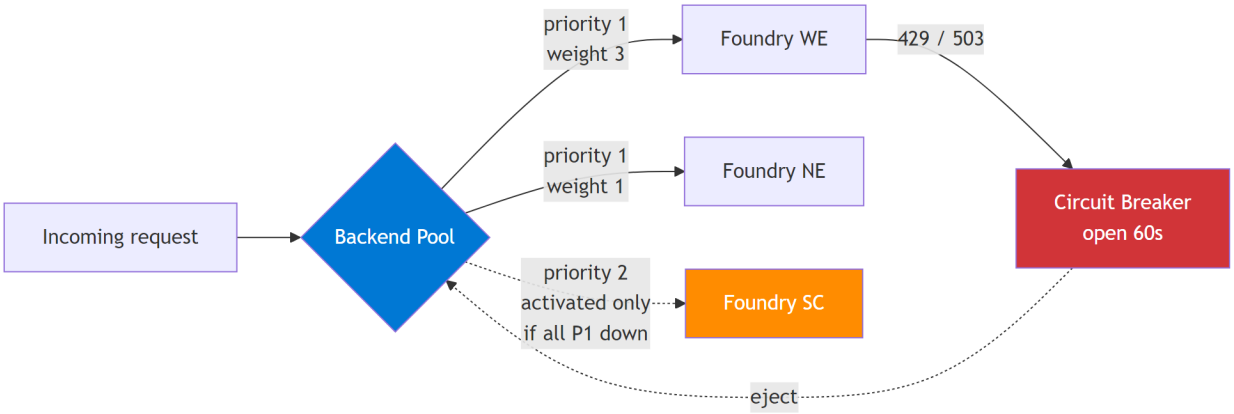

Figure 3: Diagram 3

- **Priority** — backends with priority **1** receive all traffic; priority **2** activates only when all priority-1 backends are tripped.
- **Weight** — within a priority tier, traffic is split proportionally (here `3:1` ratio between WE and NE).
- **Circuit breaker** — after N consecutive failures, the backend is taken out of rotation for a cooldown period. Native trigger on `429` and `5xx`.

5.2 Bicep — pool with circuit breaker

```
resource foundryWE 'Microsoft.ApiManagement/service/backends@2024-05-01' = {
  name: 'foundry-we'
  parent: apim
  properties: {
    url: 'https://foundry-we.services.ai.azure.com/api/projects/default'
    protocol: 'http'
    circuitBreaker: {
      rules: [
        {
          name: 'aoai-throttle'
          failureCondition: {
            count: 3
            interval: 'PT1M'
            statusCodeRanges: [ { min: 429, max: 429 }, { min: 500, max: 599
↪   } ]
          }
          tripDuration: 'PT1M'
        }
      ]
    }
  }
}
```

```
resource foundryNE 'Microsoft.ApiManagement/service/backends@2024-05-01' = {
  name: 'foundry-ne'
  parent: apim
  properties: {
    url: 'https://foundry-ne.services.ai.azure.com/api/projects/default'
    protocol: 'http'
  }
}

resource foundrySC 'Microsoft.ApiManagement/service/backends@2024-05-01' = {
  name: 'foundry-sc'
  parent: apim
  properties: {
    url: 'https://foundry-sc.services.ai.azure.com/api/projects/default'
    protocol: 'http'
  }
}

resource pool 'Microsoft.ApiManagement/service/backends@2024-05-01' = {
  name: 'foundry-pool'
  parent: apim
  properties: {
    type: 'Pool'
    pool: {
      services: [
        { id: foundryWE.id, weight: 3, priority: 1 }
        { id: foundryNE.id, weight: 1, priority: 1 }
        { id: foundrySC.id, weight: 1, priority: 2 }
      ]
    }
  }
}
```

5.3 Policy

```
<policies>
  <inbound>
    <base />
    <set-backend-service backend-id="foundry-pool" />
    <authentication-managed-identity
 ↪  resource="https://cognitiveservices.azure.com" />
  </inbound>
</policies>
```

That's it. APIM handles failover transparently — the client never sees a **429**.

---

6 Pattern 3 — Semantic Routing (Preview)

`llm-semantic-routing` (and its Azure-OpenAI-specific cousin `azure-openai-semantic-routing`) routes a request to a backend based on the **embedding similarity** between the prompt and a set of route definitions. It uses an embedding model to convert the user's prompt into a vector, then picks the route whose anchor phrase is closest in cosine distance.

6.1 Example — route by intent

```
<policies>
  <inbound>
    <base />
    <azure-openai-semantic-routing
        embeddings-backend-id="text-embedding-3-small"
        threshold="0.75">
      <route backend-id="foundry-codegen">
        <phrase>write code · refactor · debug · explain function</phrase>
      </route>
      <route backend-id="foundry-summarize">
        <phrase>summarize · TL;DR · extract key points</phrase>
      </route>
      <route backend-id="foundry-creative">
        <phrase>write a story · poem · marketing copy · brainstorm</phrase>
      </route>
      <default-route backend-id="foundry-pool" />
    </azure-openai-semantic-routing>
  </inbound>
</policies>
```

6.2 Cost & latency

Factor	Impact
Extra embedding call	`text-embedding-3-small` ≈ $0.02 / 1M tokens — negligible for short prompts
Added latency	+50 to +150 ms (one embedding round-trip per request)
Threshold tuning	Below `0.7` → too many false positives; above `0.85` → most requests fall to `<default-route>`

> **Always declare a `<default-route>`** — otherwise unmatched prompts fail with `503`. In preview, also pin a max-results-per-route to avoid pathological hot spots.

---

7 Centralized Bicep IaC for Multi-Foundry

A single declarative file drives the whole topology. Adding a new region or model becomes **a one-line change** — backends, RBAC and the pool regenerate automatically.

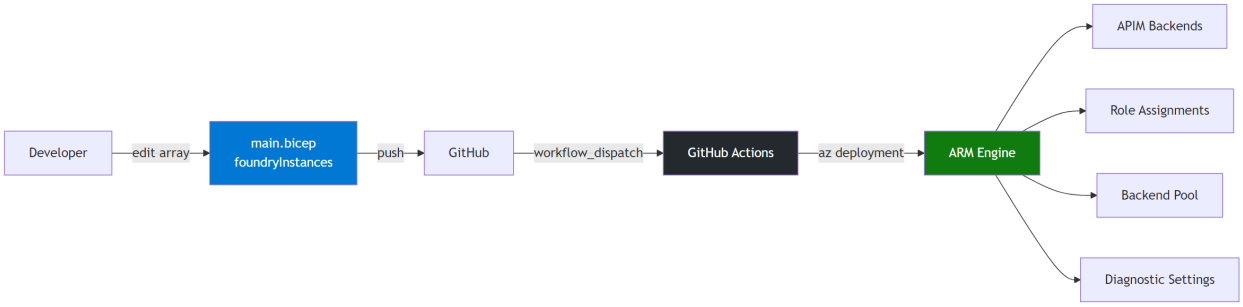

Figure 4: Diagram 4

7.1 `main.bicep` — full pattern

```
@description('Declarative list of Foundry instances behind the gateway')
param foundryInstances array = [
  { name: 'foundry-we', region: 'westeurope',     priority: 1, weight: 3 }
  { name: 'foundry-ne', region: 'northeurope',    priority: 1, weight: 1 }
  { name: 'foundry-sc', region: 'swedencentral',  priority: 2, weight: 1 }
]

resource apim 'Microsoft.ApiManagement/service@2024-05-01' existing = {
  name: 'apim-ai-gateway'
}

// 0. Foundry resources (or 'existing' if pre-provisioned)
resource foundryRes 'Microsoft.CognitiveServices/accounts@2024-10-01' = [for
 ↪  f in foundryInstances: {
  name: f.name
  location: f.region
  kind: 'AIServices'
  sku: { name: 'S0' }
  identity: { type: 'SystemAssigned' }
  properties: { customSubDomainName: f.name }
}]

// 1. Backends auto-created
resource backends 'Microsoft.ApiManagement/service/backends@2024-05-01' =
 ↪  [for (f, i) in foundryInstances: {
  name: 'be-${f.name}'
  parent: apim
  properties: {
    url: 'https://${f.name}.services.ai.azure.com/api/projects/default'
```

```
    protocol: 'http'
    circuitBreaker: {
      rules: [ {
        name: 'aoai-throttle'
        failureCondition: {
          count: 3
          interval: 'PT1M'
         statusCodeRanges: [ { min: 429, max: 429 }, { min: 500, max: 599 } ]
        }
        tripDuration: 'PT1M'
      } ]
    }
  }
}]

// 2. RBAC auto-grant — APIM MI gets 'Cognitive Services User' on each
 ↪  Foundry
var cognitiveServicesUserRoleId = 'a97b65f3-24c7-4388-baec-2e87135dc908'

resource roleAssignments
 ↪  'Microsoft.Authorization/roleAssignments@2022-04-01' = [for (f, i) in
 ↪  foundryInstances: {
  name: guid(f.name, apim.id, cognitiveServicesUserRoleId)
  scope: foundryRes[i]
  properties: {
    principalId: apim.identity.principalId
    principalType: 'ServicePrincipal'
    roleDefinitionId:
 ↪  subscriptionResourceId('Microsoft.Authorization/roleDefinitions',
 ↪  cognitiveServicesUserRoleId)
  }
}]

// 3. Backend Pool — rebuilt on every deployment
resource pool 'Microsoft.ApiManagement/service/backends@2024-05-01' = {
  name: 'foundry-pool'
  parent: apim
  properties: {
    type: 'Pool'
    pool: {
      services: [for (f, i) in foundryInstances: {
        id: backends[i].id
        weight: f.weight
        priority: f.priority
      }]
    }
  }
}
```

7.2 Add an instance — workflow

1. Append one row to `foundryInstances`.
2. `git push` → GitHub Actions triggers `az deployment group create`.
3. APIM is up to date — new backend, RBAC, pool, monitoring — **no portal click required**.

Sample reference — the full module (with policies, monitoring and KQL workbooks) is published in the official `Azure-Samples/AI-Gateway` repo.

---

8 Quota Flow — How APIM Knows a Backend Is Saturated

Azure OpenAI / Foundry returns three relevant signals on every response. APIM uses these to decide when to fail over.

Header / Status	Meaning	APIM Reaction
`x-ratelimit-remaining-tokens`	Tokens left in the current TPM window	Informational — surfaced in `azure-openai-emit-token-metric`
`x-ratelimit-remaining-requests`	Requests left in the current RPM window	Informational
`429 Too Many Requests`	Quota exceeded, request rejected	**Circuit breaker counter +1** + Foundry returns `retry-after`
`503 Service Unavailable`	Backend transient failure	**Circuit breaker counter +1**

8.1 Sequence — one request, multiple backends

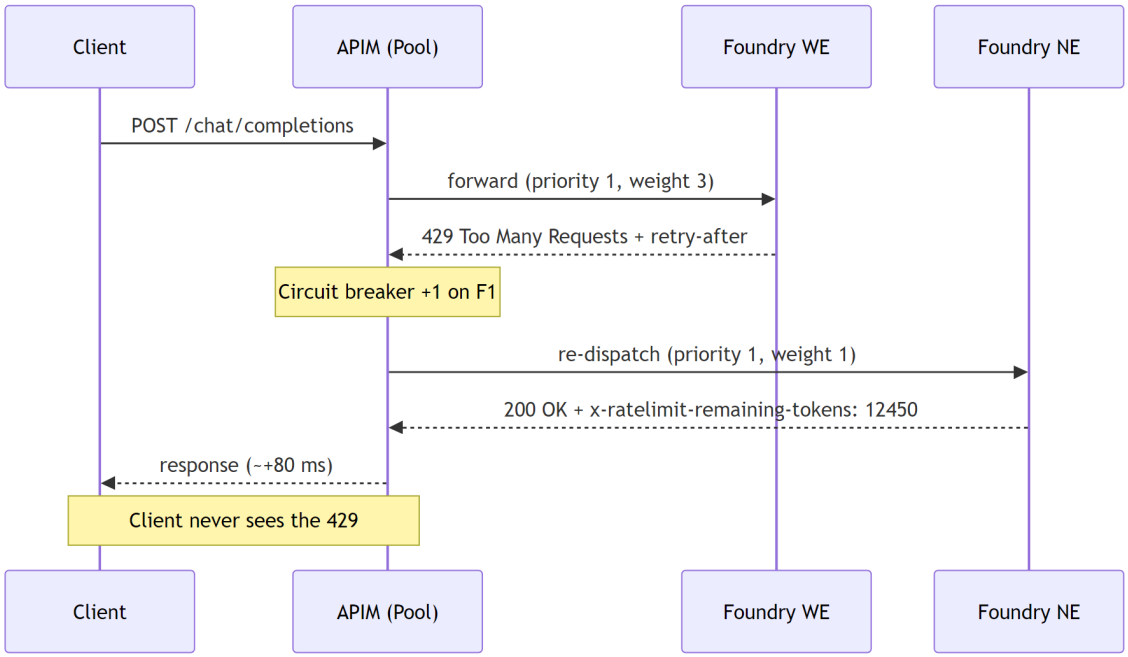

Figure 5: Diagram 5

After three `429`s in 60 seconds, F1 is **opened** for 60 s and all traffic shifts to F2 — and to F3 (priority 2) if both P1 backends are open.

---

9 Mid-Chat Throttling — What Happens During a Conversation

A conversation is **N independent requests** correlated by `thread_id` (Foundry Agents) or by `messages[]` history (raw Chat Completions). APIM re-evaluates routing at **every call** — but a stream that has already started cannot be re-routed.

9.1 Three cases

Case	When the 429 lands	APIM behavior	UX impact
Pre-stream	Before the first SSE chunk	APIM swaps backend silently	Invisible (~+80 ms)
Mid-stream	After the first chunk	Stream is **interrupted** with a network-style error	Client must retry — usually with a "continue" prompt
Multi-turn	Between turn N and N+1	Next call routed to a healthy backend	Invisible — conversation continues

9.2 Mitigation — continuation pattern

```
def safe_chat(client, messages, max_continuations=2):
    for _ in range(1 + max_continuations):
        try:
            chunks = list(client.chat_stream(messages))
            yield from chunks
            if not is_truncated(chunks): return
            messages.append({"role": "user", "content": "Continue."})
        except RateLimitError:
            time.sleep(1)
```

Foundry's **Threads API** stores conversation state server-side — when a stream fails mid-flight, you re-issue the same `thread_id` and the agent resumes at the last assistant message.

---

10 Agent-to-Agent Throttling Patterns

When agents call other agents (multi-agent orchestration), throttling no longer affects a single user request — it can **cascade** across the whole workflow. Five techniques to keep this under control:

10.1 1. Budget per conversation, not per call

```
<azure-openai-token-limit
    counter-key="@(context.Request.Headers.GetValueOrDefault("x-
↪   conversation-id","anon"))"
    tokens-per-minute="60000"
    estimate-prompt-tokens="true"
    remaining-tokens-header-name="x-tokens-remaining" />
```

A 5-agent collaboration with `counter-key="@(thread_id)"` shares the same budget — no agent can starve the others.

10.2 2. Required headers from the orchestrator

Header	Purpose
`x-conversation-id`	Token budget scope
`x-agent-id`	Per-agent metric splitting
`x-priority`	`critical` / `best-effort` — used for throttle policies
`x-correlation-id`	End-to-end tracing in App Insights

10.3 3. Best-effort vs critical

```
<choose>
  <when condition="@(context.Request.Headers.GetValueOrDefault("x-priority
  ",")=="best-effort")">
    <set-backend-service backend-id="foundry-pool-cheap" />
  </when>
  <otherwise>
    <set-backend-service backend-id="foundry-pool-premium" />
  </otherwise>
</choose>
```

Non-critical agents (logging, summarization, reranking) get cheap models; the user-facing answer agent stays on the premium pool.

10.4 4. Circuit breaker at orchestrator level

Don't only rely on APIM. The orchestrator (Semantic Kernel, AutoGen, LangGraph…) should track per-agent failure rates and **stop calling** an agent that returned `429` more than X times in T seconds — falling back to a degraded response.

10.5 5. Idempotency keys

Every A2A call carries `Idempotency-Key: <uuid>`. APIM caches the response for 5 minutes (`llm-semantic-cache-store`) so retries are free.

10.6 Sequence — A2A with degradation

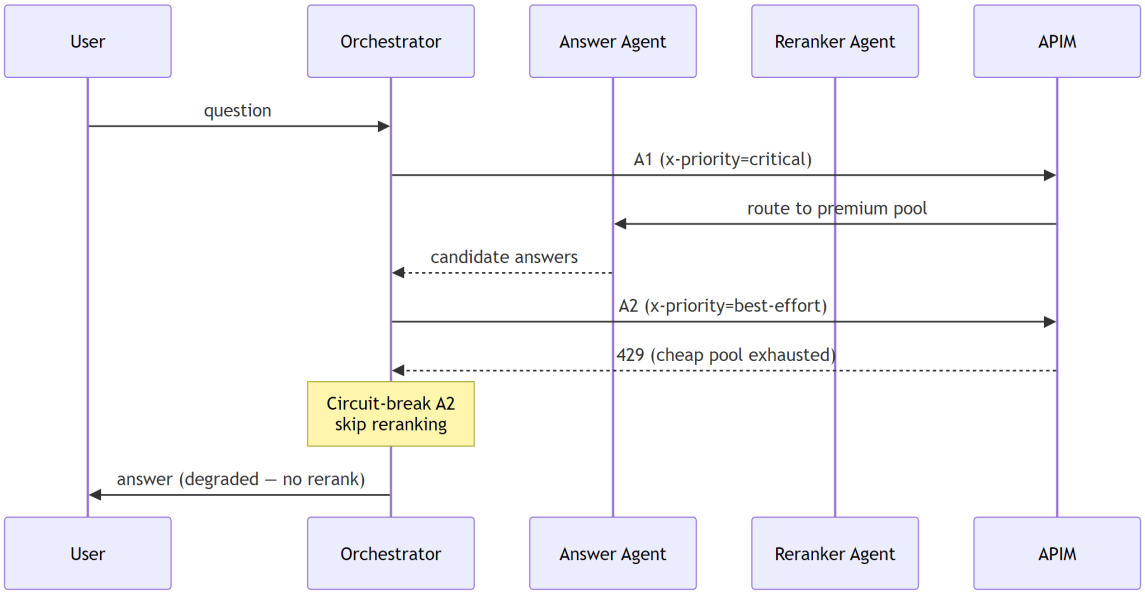

Figure 6: Diagram 6

The user gets a slightly less polished answer, but the system **never returns an error**.

---

11 Test & Simulation — Validating the Circuit Breaker

Before going to production, you must **prove** that the failover actually triggers. Real `429`s from Azure OpenAI are unpredictable and expensive to reproduce — instead, **inject** synthetic errors at the gateway, the backend, or the policy layer.

11.1 Three injection levels

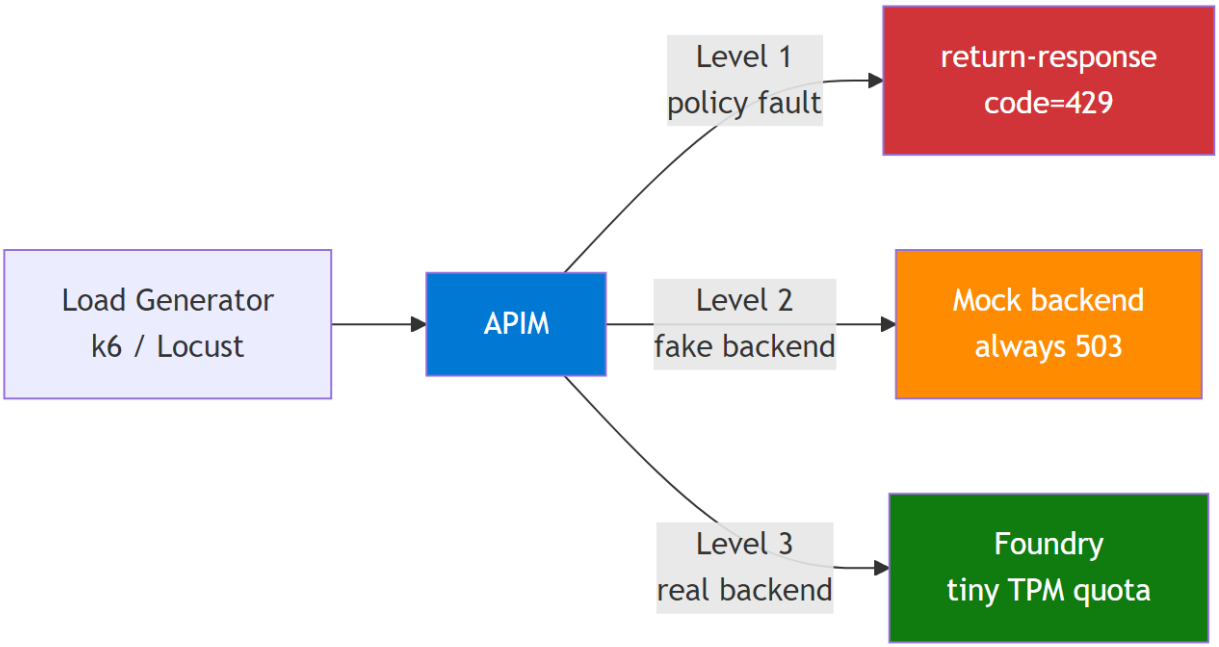

Figure 7: Diagram 7

Level	Injection point	Pros	Cons
1 — Policy fault	APIM `<return-response>` on a target backend	Instant, free, deterministic	Doesn't exercise the real network path
2 — Mock backend	Dedicated APIM backend that always returns `503` / `429`	Realistic transport, controllable	Adds a fake backend resource
3 — Real throttling	Provision a Foundry deployment with **tiny TPM** (e.g. 1k) and hammer it	Truest test	Slowest, costs real tokens

11.2 Level 1 — Inject 429 via APIM policy

Add a temporary **<choose>** block on one backend in the pool — toggled by an **x-fault** header so prod traffic is unaffected:

```
<inbound>
  <base />
  <choose>
    <when condition="@(context.Request.Headers.GetValueOrDefault("x-fault
 ↪  ","")=="429"
                    && context.Request.Url.Host.Contains("foundry-we"))">
      <return-response>
        <set-status code="429" reason="Too Many Requests" />
        <set-header name="retry-after" exists-action="override">
          <value>10</value>
        </set-header>

 ↪  <set-body>{"error":{"code":"throttle","message":"injected"}}</set-body>
      </return-response>
    </when>
  </choose>
  <set-backend-service backend-id="foundry-pool" />
</inbound>
```

11.3 Level 2 — Mock backend "always 429"

```
resource backendChaos 'Microsoft.ApiManagement/service/backends@2024-05-01'
 ↪  = {
  name: 'be-chaos-429'
  parent: apim
  properties: {
    url: 'https://httpstat.us/429'
    protocol: 'http'
  }
}

resource poolChaos 'Microsoft.ApiManagement/service/backends@2024-05-01' = {
  name: 'foundry-pool-chaos'
  parent: apim
  properties: {
    type: 'Pool'
    pool: {
      services: [
        { id: backendChaos.id, weight: 1, priority: 1 }
        { id: foundryWE.id,    weight: 1, priority: 2 }
      ]
    }
  }
```

```
}
```

Switch one API operation to **`foundry-pool-chaos`** — every call hits the broken backend first, the circuit opens, and traffic flips to the priority-2 real Foundry. Watch the tripping happen live.

11.4 Validation script (k6 + jq)

The same script proves three things in one run: (1) the circuit opens after N failures, (2) the cooldown lasts T seconds, (3) end-to-end latency stays acceptable during failover.

```
// chaos.js — run with: k6 run chaos.js
import http from 'k6/http';
import { check, sleep } from 'k6';

const APIM_URL = __ENV.APIM_URL;        // https://apim.../chat/completions
const APIM_KEY = __ENV.APIM_KEY;

export const options = {
  scenarios: {
    inject_429: {
      executor: 'constant-arrival-rate',
      rate: 20, timeUnit: '1s', duration: '90s',
      preAllocatedVUs: 30,
    },
  },
  thresholds: {
    'checks{tag:failover}':  ['rate>0.95'],   // ≥95% requests succeed
     ↪  despite injection
    'http_req_duration{tag:failover}': ['p(95)<3000'],
  },
};

export default function () {
  const res = http.post(APIM_URL,
    JSON.stringify({ messages: [{ role: 'user', content: 'ping' }],
     ↪  max_tokens: 5 }),
    { headers: {
        'Ocp-Apim-Subscription-Key': APIM_KEY,
        'Content-Type': 'application/json',
       'x-fault': '429',                      // triggers the injected fault
      },
      tags: { tag: 'failover' },
    });

  check(res, {
    'status is 200': (r) => r.status === 200,
    'served by failover backend': (r) => r.headers['X-Apim-Backend'] !==
     ↪  'foundry-we',
```

```
  });
  sleep(0.05);
}
```

Expected outcome (read in Application Insights, KQL):

```
ApiManagementGatewayLogs
| where TimeGenerated > ago(5m)
| summarize
    requests   = count(),
    success    = countif(ResponseCode == 200),
    throttled  = countif(ResponseCode == 429),
    failover   = countif(BackendId != "be-foundry-we" and ResponseCode ==
 ↪  200)
  by bin(TimeGenerated, 10s)
| render timechart
```

You should see: a burst of `429` for ~3 seconds, the circuit breaker opens, the `failover` count climbs and `success` stays > 95%.

11.5 Reset between runs

```
# Force-close the circuit by clearing the breaker state via Azure CLI
az rest --method POST --url \
  "https://management.azure.com/subscriptions/$SUB/resourceGroups/$RG
   ↪  /providers/Microsoft.ApiManagement/service/$APIM/backends/be-foundry-
   ↪  we/reconnect?api-version=2024-05-01"
```

> **Tip — chaos toolkits** — the Azure Chaos Studio supports network faults on APIM and can pause an outbound NSG rule to simulate a regional outage end-to-end, beyond what `<return-response>` covers.

---

12 Observability & Operations

What you cannot measure, you cannot govern. APIM emits four signal categories that, combined, give Ops teams full visibility into the AI Gateway.

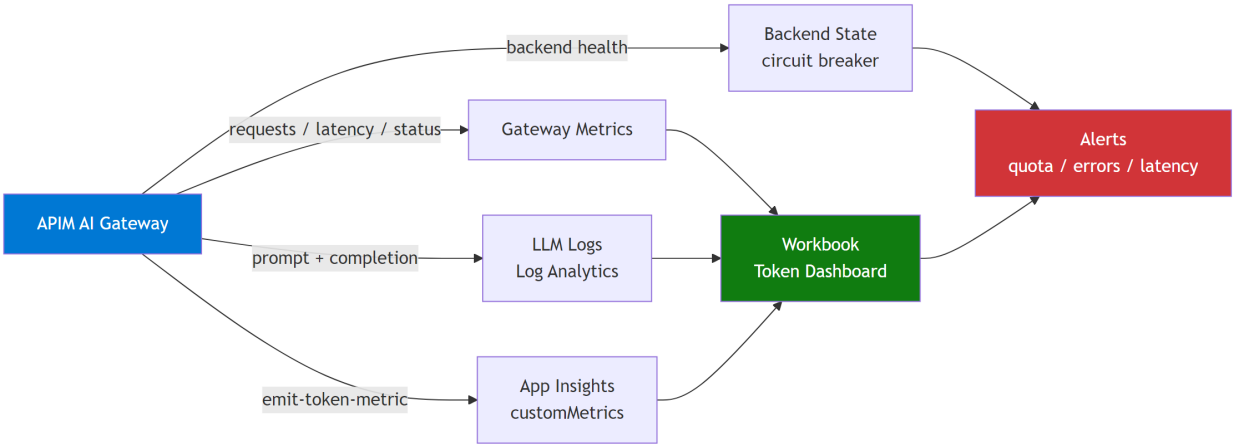

Figure 8: Diagram 8

12.1 What to enable

Signal	How	Use
LLM logs (full prompt + completion)	APIM diagnostic setting → `LLMLogs` table in Log Analytics	Audit, replay, quality analysis, internal billing
Token metrics	`azure-openai-emit-token-metric` policy — emits `prompt_tokens`, `completion_tokens`, `total_tokens` to App Insights with dimensions (subscription, user, model, region)	Per-team / per-user usage breakdown
Gateway logs	`ApiManagementGatewayLogs` — request/response code, backend id, duration	SLA monitoring, error-rate dashboards
Backend health	Backend resource state (`available` / `circuit-open`) via Azure Monitor	Page on-call when a region trips
AI Foundry traces	Foundry → App Insights connection (see companion doc)	Per-agent token + tool usage

Enable LLM logging on the API:

```
<inbound>
  <base />
  <llm-emit-token-metric namespace="ai-gateway">
```

```
    <dimension name="subscription-id" value="@(context.Subscription.Id)" />
    <dimension name="api-id"          value="@(context.Api.Id)" />
    <dimension name="user-id"         value="@(context.User.Id ?? "anon")" />
    <dimension name="model"
     ↪  value="@((string)context.Variables["model"])" />
  </llm-emit-token-metric>
</inbound>
```

12.2 Built-in workbook & alerts

APIM ships a **GenAI APIs dashboard** (Azure portal → APIM → Monitoring → Workbooks → *AI Gateway*) showing token consumption trends, top consumers, error rates, latency p50/p95/p99 — out of the box.

Recommended alerts (Azure Monitor):

Trigger	Threshold
Subscription ≥ 90 % of daily token quota	per `subscription-id` dimension
Backend error rate > 5 % over 5 min	per `BackendId`
Circuit breaker opened	any backend transitions to `circuit-open`
p95 latency > 8 s over 5 min	per `Operation`
Cost burn-rate > €X / hour	computed metric (tokens × price)

> **Cross-link** — KQL queries, retention strategy and dashboard JSON are in Foundry_Agent_Monitoring_APIM. This section adds the **multi-backend dimensions** (`BackendId`, `region`) that single-Foundry monitoring doesn't need.

---

13 Cost & FinOps

The AI Gateway adds two cost lines (APIM units + log storage) but unlocks several **savings levers** that usually pay for themselves.

13.1 Cost decomposition

Item	Order of magnitude	Lever
APIM service (Standard v2 / Premium)	from a few hundred €/month per unit, scaled by RPS	Pick the right tier — see below
Log Analytics ingestion	~ €2.30 / GB ingested	Sample LLM logs (10 % in prod), shorten retention
App Insights metrics	minimal — custom metrics are cheap	n/a

Item	Order of magnitude	Lever
Foundry tokens	the dominant line	**Semantic cache + quotas**
Egress + Private Link	low	Co-locate APIM and Foundry in the same region

13.2 Levers — how the gateway lowers your bill

13.2.1 1. Semantic cache — 30-50 % token savings on FAQ workloads

```
<inbound>
  <base />
  <llm-semantic-cache-lookup
      score-threshold="0.05"
      embeddings-backend-id="text-embedding-3-small">
    <vary-by>@(context.Subscription.Id)</vary-by>
  </llm-semantic-cache-lookup>
</inbound>
<outbound>
  <base />
  <llm-semantic-cache-store duration="3600" />
</outbound>
```

A "what's the leave policy?" prompt asked 50 ways a day → only the first call hits the model. Pricing: a cached hit costs ~ one cheap embedding call (`text-embedding-3-small` ≈ $0.02 / 1M tokens) instead of a `gpt-4o` round-trip (~ $5 / 1M input + $15 / 1M output). Break-even ≈ 1 cache hit per 250 misses.

13.2.2 2. Quotas — turn surprise bills into predictable ones

Per-team `llm-token-limit` (covered in *Security & Governance*) caps how much each Product can spend, which directly maps to a monthly budget cap.

13.2.3 3. Route cheap → expensive

Use Pattern 1 or Pattern 3 to send simple prompts to `gpt-4o-mini` (cost factor ~15× lower than `gpt-4o`) and reserve frontier models for complex requests. Even a 70/30 split typically halves the bill.

13.2.4 4. Provisioned Throughput Units (PTU) first

Mix priorities so that a flat-fee **PTU deployment** is consumed first (`priority: 1, weight: 10`), with pay-as-you-go regions as overflow (`priority: 2`). This drives PTU utilization to ~95 % — the only way PTUs are cheaper than PAYG.

```
{ id: foundryPTU.id,    weight: 10, priority: 1 }   // flat fee, use first
{ id: foundryPaygWE.id, weight: 1,  priority: 2 }   // overflow
{ id: foundryPaygNE.id, weight: 1,  priority: 2 }
```

13.2.5 5. APIM tier sizing

Tier	When
Developer	dev / sandbox only — no SLA
Standard v2	most production AI Gateway workloads up to ~ 1 000 RPS, supports VNet integration
Premium	required for multi-region APIM, availability zones, > 1 000 RPS, large enterprises

Avoid over-provisioning — APIM scales horizontally with **gateway units**, but every extra unit is €€€. Start at 1 unit, watch the *Capacity* metric, scale only when it crosses ~ 60 %.

13.3 Cost monitoring

Add a computed metric in App Insights workbook:

```
customMetrics
| where name == "total_tokens"
| extend price_per_1k =
      case(model == "gpt-4o",      0.005,
           model == "gpt-4o-mini", 0.00015,
           model startswith "text-embedding", 0.00002,
           0.0)
| extend cost_eur = (toreal(value) / 1000.0) * price_per_1k
| summarize cost = sum(cost_eur) by bin(timestamp, 1h),
 ↪  tostring(customDimensions.["subscription-id"])
| render timechart
```

Combined with an alert at 80 % of monthly budget, you get a hard FinOps guardrail.

---

14 Decision Matrix

Question	Recommended pattern
One model, one region, < 100 RPS	Direct call to Foundry — no APIM needed
Multiple models exposed via one URL	**Pattern 1** — Named backends + header routing

Question	Recommended pattern
Multi-region resilience required	**Pattern 2** — Backend Pool + circuit breaker
Cost optimization (route cheap vs frontier)	**Pattern 3** — Semantic routing (preview)
Several Foundry resources to provision and govern	**Centralized Bicep IaC** + GitHub Actions
User-facing chat with intermittent 429s	Pattern 2 + continuation prompts
Multi-agent orchestration with budget	Per-conversation `counter-key` + best-effort priority tier

---

15 References

15.1 Microsoft Learn

- Backends in API Management — types, pools, circuit breaker
- Backends — load-balanced pool
- Tutorial: load-balance Azure OpenAI traffic
- GenAI Gateway capabilities
- `azure-openai-token-limit` policy
- `azure-openai-emit-token-metric` policy
- `llm-semantic-cache-lookup` policy
- `llm-semantic-routing` policy (preview)
- `llm-content-safety` policy — prompt shield + content moderation at the gateway
- Azure AI Content Safety — categories, severity thresholds, prompt-shield
- Authenticate APIM to Azure OpenAI with managed identity
- ARM/Bicep template — `service/backends`
- Bicep loops
- Built-in role — Cognitive Services User
- Azure AI Foundry RBAC
- Foundry quotas, limits & regions

15.2 AI Gateway design patterns

- Designing your AI Gateway with APIM — Tech Community reference design
- Smart load-balancing for OpenAI endpoints with APIM — original pattern that pre-dated the Pool feature
- GenAI Gateway capabilities — overview
- Azure AI Gateway architecture (CAF) — Cloud Adoption Framework reference

15.3 Network injection & private connectivity

- Use a virtual network with API Management — Internal vs External VNet modes

- Inject APIM into a VNet — internal mode
- Connect privately to API Management with Private Link — inbound private endpoint
- Configure outbound access through Private Endpoint — APIM → Foundry over a private link
- APIM v2 networking model (stv2) — required for newest features
- Foundry / AI Services with Private Endpoint — locking down the Foundry side
- DNS for Private Link in Hub-Spoke — the common pitfall

15.4 Security best practices

- APIM security baseline (Microsoft Cloud Security Benchmark)
- Mitigate OWASP API Top-10 with APIM
- Landing Zone accelerator for APIM
- Defender for APIs — runtime threat detection
- Validate JWT in APIM policies
- Managed identities in APIM — no keys for backends

15.5 Resilience & chaos testing

- Azure Chaos Studio — fault injection service
- Resilience patterns for AI workloads — quota & failover guidance
- k6 — load testing toolkit — used in the Test & Simulation chapter

15.6 Sample repositories

- `Azure-Samples/AI-Gateway` — full Bicep modules for backend pools, semantic routing, monitoring
  - `labs/backend-pool-load-balancing/`
  - `labs/backend-pool-circuit-breaker/`
  - `labs/semantic-routing/`

15.7 Companion document

- Foundry_Agent_Monitoring_APIM — single-Foundry per-user token tracking, KQL queries, dashboards